

SE 4485: Software Engineering Projects

Fall 2026

Requirement Documentation

Group Number	9
Project Title	Skills Catalog & Inventory (Team A)
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Skills Catalog & Inventory

A. ABSTRACT

This is the Requirements Document for the Skills Catalog & Inventory project sponsored by Sabre Corporation. This project is one of the ten projects for this course SE 4485.001 at the University of Texas at Dallas. This document complies with the Software Engineering Standards set by the IEEE Standard for Software Project Management Plans (IEEE Std 1058-1998). The document sets the functional and non-functional requirements for the project.

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1. Introduction

1.1. Overview

The purpose of this project is to build a cloud-based software tool to help organizations overcome the challenge of tracking employee skills across complex team structures. To make workforce capabilities clear and connected, the platform links three core areas: the Skills Catalog/Master Data (standardized definitions and hierarchy of all skills), Organization Structure (company's teams, roles, and relationships), and Skills Inventory (mapping employees to skills).

1.2. Purpose and scope of the document

The purpose of this document is to define the functional and structural requirements for developing the cloud-based software. As the semester goes on, we will explore and choose what our tech stack would be to satisfy the requirements. The scope of the document is as follows:

1. Core Data Architecture: Designing the relationship between Master Data, Organization Structure, and Operational Data
2. Stakeholder Groups:
 - a. Workforce Planners - Build & maintain the Skills catalog
 - b. Resource Managers - Enter, view, and update Skills per resource, creating and maintaining the Skills Inventory
 - c. Organizational Leaders - Skills inventory summary views to manage the existing organization and plan future workforce strategies
 - d. Business Operations - Load files of additional Skills into the Skills catalog and of additional skills per resource into the Skills Inventory

1.3. Brief overview of the product

A cloud-based software tool has to be designed and built to create and maintain a Skills Catalog. The tool has to be built with a focus on an exceptional user interface (UI) with the ability to enter, view, and update skills per resource, thereby creating the Skills Inventory. The purpose of this tool is to simplify the process of keeping track of which resources in the company have which skills and create a skills catalog that users can easily view, understand, and update. In the organization, leaders view summaries of skill inventory to manage organizational skills capacity and plan future workplace strategies.

1.4 Description of the structure of the document

This Project Management Plan provides a structured roadmap for the development of the Skills Catalog & Inventory system. The document is organized into the following sections:

1. Introduction: Defines the project's purpose, scope, and product overview.
2. Use Case Model for Functional Requirements: Describes the functional requirements using graphical models and detailed textual descriptions for each use case.
3. Rationale for Your Use Case Model: Provides the justification and reasoning behind our chosen use case structure.
4. Non-Functional Requirements: Specifies quality attributes, system constraints, performance requirements, and technical standards.
5. Professional Standards: Sets expectations for team conduct and deliverables.
6. Evidence of Configuration Management: Links to relevant repository management tools.
7. Engineering Standards and Multiple Constraints: Lists the industry standards guiding the project.
8. Additional References & Appendix: Includes supplementary material and the professional standards guideline.

2. Use Case Model for Functional Requirements

2.1 Graphic Use Case Model

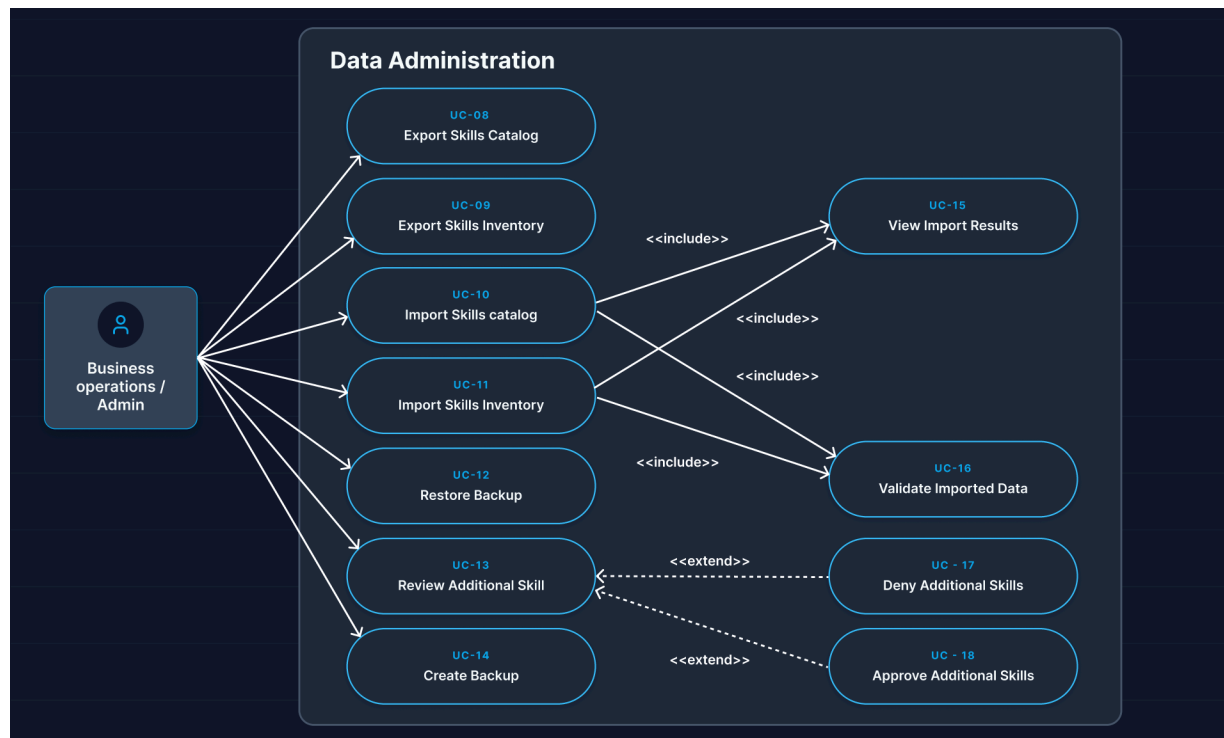


Figure 1: Use Case Model for Data Administration

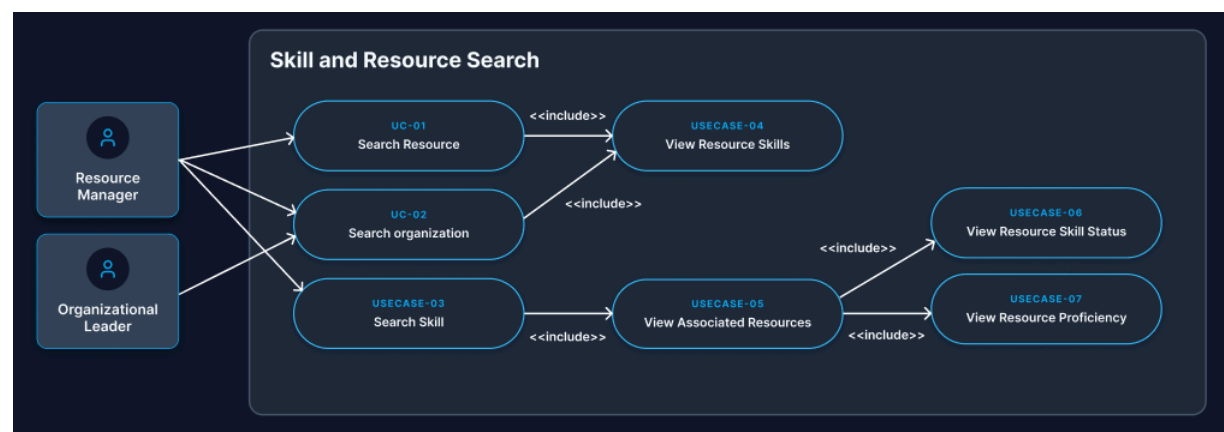


Figure 2: Use Case Model for Skill and Resource Search

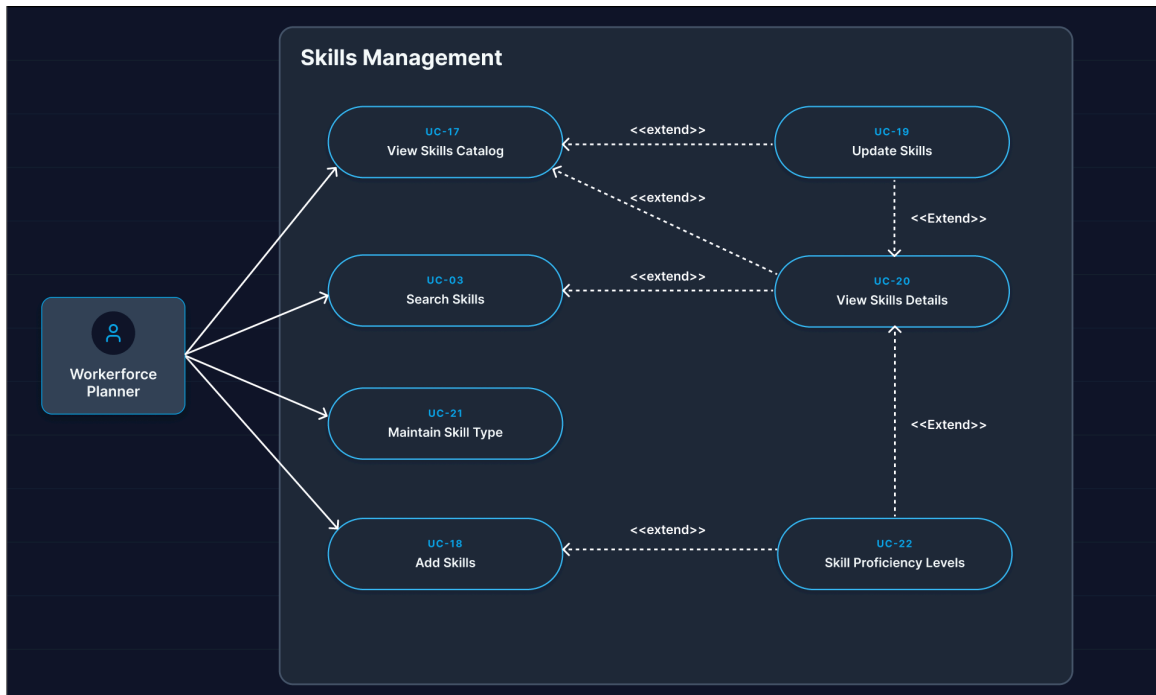


Figure 3: Use Case Model for Workforce Planner

2.2 Textual Description

General / Business Operations Functional Requirements	Description
Import Skills Catalog	The system shall allow admins to import skill catalog data from a file
Load additional skills	The system shall allow admins to load additional skills into the existing skills catalog
Validate Imported Data	The system shall validate all imported data before adding it to the database
View Import Errors/Results	The system shall allow the viewing of errors and results of imports
Export Data	The system shall allow the exportation of the skills catalog and inventory to a file
Backup	The system shall allow the creation of a backup of system data
Restoration	The system shall allow admins to restore data from a previously created backup
Approve or deny additional skills	The system shall allow admins to either approve or deny when new skill is added to the catalog
Web-based system	The system shall be web-based and accessible from the cloud
Cancel	The system shall allow the cancellation of association between resource and skill

Workforce Planner Functional Requirements	Description
Views Skills Catalog	The system shall allow Workforce Planners to view the Skills Catalog and its existing skills.
Add Skills	The system shall allow Workforce Planners to add new skills to the Skills Catalog.
Update Skills	The system shall allow Workforce Planners to update existing skill information.
Search Skills	The system shall allow Workforce Planners to search for a skill using full or partial skill names.
View Skills Details	The system shall allow Workforce Planners to select a skill to view its details such as name, type, and status of that skill.
Maintain Skill Type	The system shall support maintaining skill types.
Skill Proficiency Levels	Assign and update proficiency levels (e.g., Beginner, Intermediate, Expert) for skills assigned to resources.
Graphical UI	The system shall provide graphical UI components to help navigate and maintain complex skill data.
AI assistance	The system may provide AI-assisted functionality to maintain an efficient skills catalog.

Resource Manager Functional Requirements	Description
Creating/Maintaining Skills Inventory	The system shall allow Resource Managers to enter, view, update, and remove skills per person (resource).
Search for a specific resource or organization to view their skills	The system shall allow Resource Managers to search for a specific resource or specific organization and its resources to view their skills.
Search for a specific skill and see all associated resources	The system shall allow Resource Managers to search for a specific skill and display all resources associated with that skill along with their proficiency and status.
View UI	The system shall allow Resource Managers to access a UI focused on specific resource(s) within the organization to enter, view, and update skills per resource.
Create, view, and maintain association between skill and resource	The system shall allow Resource Managers to create, view, update, and remove associations between skills and individual resources
View specific Skill UI	The system shall allow Resource Managers to access a UI focused on skill(s) being viewed within the Skills Catalog.

Skill Validation	The system shall allow Resource Managers to validate all skill assignments and updates against predefined entries in the Master Skills Catalog.
Certification Tracking	The system shall allow Resource Managers to track certification details and verification status per resource.

Organizational Leader Functional Requirements	Description
View summary	The system shall allow OLs to view a summary of all skills
Filter Skills	The system shall allow OLs to filter the skills inventory information
View distribution	The system shall allow OLs to view a department-based distribution of skills
Graphical views	The system shall provide graphical visualization of skills inventory based on relevant filters
Skill capacity	The system shall allow OLs to identify areas of skill capacity within departments or the entire inventory
Skill Trend & Historical Analytics	The system shall allow OLs to track skill growth or decline over time across quarters or years.
Organization Hierarchy	The system shall allow OLs to view the organization hierarchy and reporting structure (who reports to whom, except for the CEO)

Use Case 01	
Use Case Name	Search Resource
Participating Actors	Resource Manager
Entry Condition(s)	Resource Manager is authenticated Organization/resource data is available
Normal Flow of Events	Resource Manager opens resource Search Resource Manager enters search criteria System searches available resources System displays matching resources Resource Manager selects a resource System displays the resource's skills
Exit Condition(s)	Matching resource(s) and their skills displayed
Exceptions	No matching resources are found Search criteria are invalid Resource data is unavailable
Special Requirements	Should support searching for a specific resource.
Authorized Role(s)	Resource Manager

Use Case 02	
Use Case Name	Search Organization
Participating Actors	Resource Manager Organizational Leader
Entry Condition(s)	Resource Manager is authenticated Organization/resource data is available
Normal Flow of Events	Resource Manager opens organization search Resource Manager enters search criteria System searches organizational structures System displays matching organizations Resource Manager selects an organization System displays resources in the organization System displays skills of selected resources
Exit Condition(s)	Organization and associated resources are displayed
Exceptions	Organization cannot be found Organization has no resources
Special Requirements	Ensure organization relationships are represented clearly
Authorized Role(s)	Resource Manager Organizational Leader

Use Case 03	
Use Case Name	Search Skills
Participating Actors	Workforce Planner Resource Manager
Entry Condition(s)	User is authenticated Skills catalog is available
Normal Flow of Events	User opens the skills search interface User enters full or partial skill name System searches skills catalog System identifies matching skills System displays the matching results User selects a result if additional information is required
Exit Condition(s)	Matching skills are displayed
Exceptions	Search returns no results Search term is empty
Special Requirements	Support full and partial names Presented through graphical UI

Authorized Role(s)	Resource Manager Workforce Planner Administrator
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Use Case 04	
Use Case Name	View Resource Skills
Participating Actors	Organizational Leader Resource Manager
Entry Condition(s)	User is authenticated Skills catalog is available
Normal Flow of Events	User selects a resource System retrieves the resource's skill associations and each skill's proficiency System displays the resource's skills User reviews information
Exit Condition(s)	Skills associated with selected resource are displayed
Exceptions	Resource cannot be found Resource has no associated skills Inventory cannot be retrieved
Special Requirements	UI should focus on selected resource(s) Skill information should be repeated
Authorized Role(s)	Resource Manager, Organizational leader

Use Case 05, 06, 07	
Use Case Name	View associated resources, proficiencies, and statuses
Participating Actors	Organizational leader Resource Manager
Entry Condition(s)	Skill is selected Skill inventory is available
Normal Flow of Events	User opens skill interface System searches for all resources with the selected skill System displays resource and associated level of proficiency and status
Exit Condition(s)	Associated resources, proficiency, and status are displayed
Exceptions	No resources associated with the skill Inventory data cannot be retrieved
Special Requirements	Results must include associated resource, proficiency, and status
Authorized Role(s)	Resource Manager Organizational Leader

Use Case 08,09	
Use Case Name	Export Skills Catalog and Skills Inventory
Participating Actors	Business operations
Entry Condition(s)	User is authenticated Exportable skills catalog and inventory data exist
Normal Flow of Events	User selects export User selects the data to export System retrieves the requested data System generates an export file System provides the file to the user
Exit Condition(s)	Requested data has been exported
Exceptions	No data exists Export fails User cancels export
Special Requirements	System must support export of entire Skills catalog and Skills inventory data
Authorized Role(s)	Business Operations / Administrator

Use Case 10	
Use Case Name	Import Skills Catalog
Participating Actors	Business operations
Entry Condition(s)	User is authenticated Import file is available
Normal Flow of Events	User selects skills catalog import User uploads the file System reads the file System validates the imported data System identifies valid and invalid records System adds valid records to the skills catalog System generates an import result System displays the result
Exit Condition(s)	Valid catalog data has been added to the skills catalog data
Exceptions	File format is invalid File cannot be read Validation fails Duplicate data is detected
Special Requirements	All imported data must be validated before being added to the database
Authorized Role(s)	Business Operations / Administrator

Use Case 11	
Use Case Name	Import Skills Inventory
Participating Actors	Business operations
Entry Condition(s)	Import file is available Resources and skills referenced by the file can be identified
Normal Flow of Events	User selects skills inventory import User uploads the file System reads the file System validates the imported data System checks referenced resources and skills System adds valid inventory associations System displays import results
Exit Condition(s)	Valid inventory information has been added
Exceptions	File format is invalid Resource or skill cannot be found Validation fails Duplicate association exists
Special Requirements	Imported skill assignments should correspond to the master skills catalog
Authorized Role(s)	Business operations / Administrator

Use Case 12	
Use Case Name	Restore Backup
Participating Actors	Business operations
Entry Condition(s)	User is authenticated At least one valid backup exists
Normal Flow of Events	Administrator selects restore System displays available backups Administrator selects a backup System requests confirmation System restores the selected data System confirms completion
Exit Condition(s)	System data has been restored from the selected backup
Exceptions	Backup is unavailable Backup is corrupted Restoration fails
Special Requirements	
Authorized Role(s)	Business Operations / Administrator

Use Case 13	
Use Case Name	Review Additional Skill
Participating Actors	Business operations
Entry Condition(s)	A new skill has been submitted for approval
Normal Flow of Events	Administrator opens pending skill submissions System displays submitted skill information Administrator reviews the proposed skill Administrator chooses to approve or deny the skill
Exit Condition(s)	Skill has been approved or denied
Exception(s)	No pending skills exist Skill information is incomplete Submission cannot be retrieved
Special Requirements	Approval status must be recorded
Authorized Role(s)	Business Operations / Administrator

Use Case 14	
Use Case Name	Create Backup
Participating Actors	Business operation
Entry Conditions (s)	The user is authenticated. System data is available for backup.
Normal Flow of Events	Business Operations selects the backup option System prepares the Skills Catalog and Skills Inventory, and related data for backup System creates the backup System stores the backup System announces that the backup was successfully created.
Exit Conditions(s)	A backup of the system data has been created and stored.
Exception(s)	Backup creation fails Data can not be accessed Storage is full
Special Requirements	The backup must preserve the data required to restore the Skills Catalog and Skills Inventory.
Authorized Role(s)	Business Operations

Use Case 15	
Use Case Name	View Import Results
Participating Actors	Business Operations / Administrator
Entry Condition(s)	Skills Catalog and Skills Inventory are imported
Normal Flow of Events	System completes validation of the imported file System compiles results summary showing valid and invalid records System displays results to user User reviews which records were successfully added or rejected
Exit Condition(s)	Import results have been displayed to user
Exceptions	No results available to display Results fail to load or display correctly
Special Requirements	Clearly separate successful imports from failed/rejected records in display
Authorized Role(s)	Business Operations / Administrator

Use Case 16	
Use Case Name	Validate Imported Data
Participating Actors	Business Operations / Administrator
Entry Condition(s)	The user is authenticated. A file has been uploaded for import into either the Skills Catalog or Skills Inventory.
Normal Flow of Events	System reads uploaded file System checks record against existing data System identifies valid records System identifies invalid or conflicting records System prepares report for user
Exit Condition(s)	Imported data is validated and separated into valid and invalid records
Exceptions	File format is unreadable, or corrupted No records are in the file Record references skill not in Catalog Record is missing required fields
Special Requirements	Clearly distinguishes valid records from invalid ones before committing data
Authorized Role(s)	Business Operations / Administrator

Use Case 17	
Use Case Name	View Skills Catalog
Participating Actors	Workforce Planner
Entry Condition(s)	Workforce Planner is authenticated, and Skills Catalog is available.
Normal Flow of Events	Workforce Planner opens the Skills Catalog. System retrieves the existing skills from the Skills Catalog. System displays existing skills. Workforce Planner reviews the Skills Catalog. Workforce Planner may select a skill.
Exit Condition(s)	The Skills Catalog and its existing skills are displayed.
Exceptions	The Skills Catalog and the existing skills are displayed.
Special Requirements	The Skills Catalog should be displayed with a clear graphical UI.
Authorized Role(s)	Workforce Planner

Use Case 18	
Use Case Name	Add Skills
Participating Actors	Workforce Planner
Entry Condition(s)	Workforce Planner is authenticated
Normal Flow of Events	Workforce Planner selects the option to add a new skill Workforce Planner enters the skill details (name, category, description, and type) System validates the entered skill details and checks for duplicate entries System saves the new skill to the Skills Catalog System displays a success confirmation message
Exit Condition(s)	New skill is successfully added to the Skills Catalog
Exceptions	Skill already exists in the Skills Catalog Required skill information is missing or invalid Workforce Planner cancels the operation
Special Requirements	System must prevent duplicate skill entries in the catalog Changes must be approved by administration
Authorized Role(s)	Workforce Planner

Use Case 19	
Use Case Name	Update Skills
Participating Actors	Workforce Planner
Entry Condition(s)	Workforce Planner is authenticated and skills catalog is available
Normal Flow of Events	Workforce Planner opens and views skills catalog System displays existing skills Workforce Planner reviews existing skills Workforce Planner extends skills catalog to update skills Workforce Planner edits skill details to maintain catalog System validates changes System saves the changes to skills catalog System displays success confirmation message
Exit Condition(s)	Changes to skill details are saved
Exceptions	Workforce Planner cancels the changes The skill does not exist in the skills catalog
Special Requirements	Changes must be approved by administration
Authorized Role(s)	Workforce Planner

Use Case 20	
Use Case Name	View Skills Catalog
Participating Actors	Workforce Planner
Entry Condition(s)	Workforce Planner is authenticated
Normal Flow of Events	Workforce Planner opens the Skills Catalog Workforce Planner retrieves skills from the Skills Catalog System display skills from Skills Catalog Workforce Planner reviews the Skills Catalog
Exit Condition(s)	Existing skills from the Skills Catalog are displayed
Exceptions	Skills Catalog is empty Skills Catalog data cannot be retrieved
Special Requirements	Must present skills using a graphical UI
Authorized Role(s)	Workforce Planner

Use Case 21	
Use Case Name	View Skills Details
Participating Actors	Workforce Planner
Entry Condition(s)	Workforce Planner is authenticated and Skills Catalog is available.
Normal Flow of Events	Workforce Planner opens the Skills Catalog Workforce Planner selects a specific skill System retrieves skill details (such as name, type, status, and description) System displays the skill's details Workforce Planner reviews the skill details
Exit Condition(s)	Detailed information for the selected skill is displayed.
Exceptions	Skill cannot be found Skill details data cannot be retrieved
Special Requirements	Must present skill details using a graphical UI.
Authorized Role(s)	Workforce Planner

Use Case 22	
Use Case Name	View Skill Proficiency Levels
Participating Actors	Workforce Planner
Entry Condition(s)	Workforce Planner is authenticated and Skills Catalog is available
Normal Flow of Events	Workforce Planner opens the Skills Catalog Workforce Planner selects a specific skill Workforce Planner requests to view proficiency levels for the skill System retrieves defined proficiency levels (e.g., Beginner, Intermediate, Expert) System displays the skill proficiency levels
Exit Condition(s)	Proficiency levels associated with the selected skill are displayed
Exceptions	Skill cannot be found Proficiency levels for the skill are not defined
Special Requirements	Must present proficiency levels using a graphical UI
Authorized Role(s)	Workforce Planner

3. Rationale For Use Case Models

The use case model is organized around the stakeholder groups — Workforce Planners, Resource Managers, Organizational Leaders, and Business Operations. Each group interacts with the Skills Catalog and Skills Inventory for different purposes.

Workforce Planners build and maintain the Skills Catalog. This includes viewing the catalog, adding skills, updating skills, and searching for skills details. The relationships are extend as viewing details, updating skills, and setting proficiency levels are optional and not steps that occur every time. UC-19 (Update Skills) extends UC-17 (View Skills Catalog) as a planner can optionally update a skill when viewing a catalog. UC-20 (View Skills Details) extends UC-03 (Search Skills) since full details only display if the Planner chooses to. UC-19 (Update Skills) extends UC-20 (View Skills Details) since a Planner can choose to update skills after viewing details. UC-22 (Skill Proficiency Levels) extends UC-20 (View Skills Details) and UC-18 (Add Skills) since setting proficiency levels is an optional step. UC-21 (Maintain Skill Type) is standalone as it is not a step of search, add, or update.

Resource Managers connect resources to skills by assigning, updating, and removing skills. Organizational Leaders review summaries, filters, and trends related to workforce planning. Since the Resource Managers and Organizational Leaders need to search for and view resource and skill data, an include was used to group shared search and view behavior — UC-01 (Search Resource) and UC-02 (Search Organization) includes UC-04 (View Resource Skills), and UC-03 (Search Skills) includes UC-05 (View Associated Resources) — rather than repeating across use cases. UC-05 (View Associated Resources) also include UC-06 (View Resource Skill Status) and UC-07 (View Resource Proficiency) since status and proficiency are typically together in the resource-skill relationship.

Business Operations handle administrator tasks, such as importing and exporting of files. They are grouped under a Data Administration model. UC-10 (Import Skills Catalog) and UC-11 (Import Skills Inventory) includes UC-15 (View Import Results) and UC-16 (Validate Imported Data) as an import needs to be validated and its results shown. UC-13 (Review Additional Skill) has an extend relationship to UC-17 (Deny Additional Skills) and UC-18 (Approve Additional Skills) since reviewing a skill is conditionally denied or approved. Exports (UC-08, UC-09) and Backups (UC-12, UC-14) are standalone since it does not use any validation steps.

4. Non-Functional Requirements

NFR	Description
Reliability of the system	The system must be able to recover from hardware or software issues in five minutes
Speed of the system	The system must respond to user input in less than 500 milliseconds under normal load.
Maintainability of the system	All updates and feature patches must deploy seamlessly into the program with zero downtime for the users.
Scalability of the system	The system must be able to support at least 100,000 users simultaneously without degradation of speed or usability of the system.
Security of the system	The system must protect organizational and resource data by restricting system functionality based on authorized user roles.
Usability of the system	The system must provide an user friendly interface that allows the user to navigate, search, view, and maintain the skills catalog easily
Availability of the system	The system must maintain an uptime of at least 99% during normal business hours, except during scheduled maintenance periods.
Accessibility of the system	The UI must comply with standard ADA regulations to ensure that the UI accounts with individuals with disabilities.
Environment Requirements	The system shall operate as a web-based system in a cloud environment.

5. Configuration Management

- **Github Repository:** <https://github.com/kimchau123/SE-4385-Group-9>

Name	Version Before	Version After	Difference between 2 consecutive versions	Review	Comments
Tobenna	0.00	0.01	Document Created	Shipped	Start of our document
Tobenna	0.01	0.02	Template Added to Document	Shipped	
Russel	0.02	0.03	Abstract	Shipped	
Tobenna	0.03	0.04	Introduction	Shipped	
Team	0.04	0.05	Configuration Management	Shipped	
Team	0.06	0.07	Define Functional Requirements	Shipped	
Team	0.07	0.08	Graphical Use Case Model for FRs	Shipped	
Team	0.08	0.09	Textual Description for each Use Case	Shipped	FRs complete
Team	0.09	0.10	Non-functional Requirements	Shipped	
Team	0.10	0.11	Professional Standards	Shipped	
Team	0.11	0.12	ENGINEERING STANDARDS AND MULTIPLE CONSTRAINTS	Shipped	
Team	0.12	0.13	ADDITIONAL REFERENCES	Shipped	
Team	0.13	0.14	Formatting the Document	Shipped	End of Our Document

6. Engineering Standards and Multiple Constraints

- IEEE Std 830-1998: Software Requirements [[pdf](#)]
- IEEE Std 29148: Requirements Engineering [[pdf](#)]
- ISO/IEC/IEEE Std 29148-2018: Systems and Software Engineering
 - Life Cycle Processes
 - Requirements Engineering [[pdf](#)]

Students should work with their project sponsor(s) to identify all the standards and constraints that should be applied for preparing this document. Additional materials that are not listed above can be included.

7. Glossary

- a. **Resources:** Individual employees or personnel within the organization
- b. **Skills Catalog (Master Data):** A centralized, standardized repository defining all skills, categories, types, and proficiency hierarchies across the organization.
- c. **Skills Inventory:** The database mapping individual resources to their specific skills, proficiency levels, and verification status.
- d. **Organization Structure / Hierarchy:** The structural representation of company teams, roles, departments, and reporting relationships ("who reports to whom").
- e. **Workforce Planners:** Stakeholders responsible for creating, building, and maintaining the Skills Catalog and skill definitions.
- f. **Resource Managers:** Stakeholders responsible for managing resource-skill associations, updating individual skills inventory, and tracking certifications.
- g. **Organizational Leaders:** Executive and managerial stakeholders who view summary analytics, skill capacity distributions, and workforce trends.
- h. **Business Operations / Admin:** System administrators responsible for data imports/exports, backup and restoration, data validation, and reviewing new skill requests.
- i. **Skill Proficiency Levels:** Standardized capability tiers (e.g., Beginner, Intermediate, Expert) assigned to resources for specific skills.

8. Additional References

- Lamsweerde, A.V., 2009. *Requirements Engineering: From System Goals to UML Models to Software Specifications*. John Wiley

Each group may include other related references that are not listed above.